

Worksheet

Numbers and indices

1 Work out each of the following without using a calculator.

a 2^6 _____ b 3^4 _____ c 7^3 _____

d $\sqrt{49}$ _____ e $\sqrt{121}$ _____ f $\sqrt{900}$ _____

2 Use a calculator to find each of the following.

a 15^3 _____ b 21^2 _____ c 34^4 _____

d $\sqrt{48}$ _____ e $\sqrt{213}$ _____ f $\sqrt{234\,000}$ _____

3 Work out each of the following without using a calculator.

a 14^3 _____ b 9^0 _____ c 4^{-2} _____

d $\sqrt[3]{216}$ _____ e $\sqrt[4]{81}$ _____ f $\sqrt{250\,000}$ _____

4 Simplify each of the following and leave your answer in index form.

a $14^8 \times 14^7$ _____ b $20^{12} \div 20^4$ _____ c $(3 \times 7)^6$ _____

d $5^3 \times 5^6$ _____ e $14^6 \div 14^2$ _____ f $(13 \times 11)^8$ _____

5 Use a calculator to find each of the following.

a 15^{-2} _____ b $\sqrt[3]{9}$ _____ c 4.8^3 _____

d $\sqrt{13.96}$ _____ e $\sqrt[5]{424}$ _____ f 6.27^{-3} _____

6 Find each of the following.

a $6^{1.5}$ _____ b $8^{2.3}$ _____ c $6^{-3.2}$ _____

7 Simplify each of the following and leave your answers in index form.

a $27^6 \times 27^9$ _____ b $18^9 \div 18^5$ _____ c $(4 \times 9)^7$ _____

d $11^7 \times 11^5$ _____ e $23^{15} \div 23^8$ _____ f $(16 \times 7)^6$ _____

8 Use a calculator to find each of the following.

a 19^{-3} _____ b $\sqrt[3]{25}$ _____ c 7.8^3 _____

d $\sqrt{27.92}$ _____ e $\sqrt[4]{244}$ _____ f 11.78^{-3} _____

Answers

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|----------------------|------------------|-------------------------------------|--------------------|-----------------|---------------------------------------|
| 1 a 64 | b 81 | c 343 | d 7 | e 11 | f 30 |
| 2 a 3375 | b 441 | c 1 336 336 | d 6.928 | e 14.595 | f 483.735 |
| 3 a 2744 | b 1 | c $\frac{1}{16}$ | d 6 | e 3 | f 500 |
| 4 a 14^{15} | b 20^8 | c $3^6 \times 7^6$ | d 5^9 | e 14^4 | f $13^8 \times 11^8$ |
| 5 a 0.004 44 | b 2.080 | c 110.592 | d 3.736 | e 3.353 | f 0.004 06 |
| 6 a 14.697 | b 119.428 | c 0.003 (or 0.003 235) | | | |
| 7 a 27^{15} | b 18^4 | c $4^7 \times 9^7$ or 36^7 | d 11^{12} | e 23^7 | f $16^6 \times 7^6$ or 112^6 |
| 8 a 0.000 146 | b 2.924 | c 474.552 | d 5.284 | e 3.952 | f 0.000 612 |